

reBot-DevArm B601-DM

PRELIMINARY SAFETY CASE — EXECUTIVE SUMMARY

6-DOF cobot arm · 4.5 kg · 650 mm reach · 1.5 kg payload · Doc ID REBOT-SC-CP-001 r0.1 · 2026-05-04 · github.com/jherrodthomas/reBot-DevArm

Important — Concept-phase safety case. Does NOT certify the arm safe for collaborative deployment. Upstream is mature on hardware and software but does NOT publish HARA, FMEA, ISO 13849-2 validation, or ISO/TS 15066 PFL measurements. Provisional acceptance: OE-1 (lab/desk supervised, fixtured) only.

Hazards	Safety functions	Standards	Open gaps	Very-high gaps	Provisional accept
33	31	6 + 4 ref	28	8	OE-1 only

Standards framework applied

Standard	Coverage
ISO 12100:2010	Master risk-assessment method
ISO 10218-1/-2:2011	Industrial robot safety (primary)
ISO/TS 15066:2016	Collaborative robot — PFL data
ISO 13849-1:2023 / -2	Safety-related parts of control systems
ISO 9001:2015	Quality management
IEC 60204-1:2018	Electrical equipment of machines
EN 61000 / ISO 9283 / 13854/57 (ref.)	EMC / robot performance / safety distances

Risk profile by operational envelope

Envelope	Description	Risk	Provisional acceptance
OE-1 Lab / desk	Supervised, fixtured, no bystanders	LOW	PROVISIONAL YES
OE-2 Education	Classroom, students operating	MEDIUM	NOT YET — 9 gaps
OE-3 Industrial cobot	Trained users, defined cell	MEDIUM	NOT YET — 14 gaps
OE-4 Public-facing	Demo with bystanders / children	HIGH	NOT YET — 17 gaps

Top residual hazards (after concept-level safety functions)

Hazard	Scenario	PLr	Residual	Reason
HZ-C06	LeRobot policy edge case	e	HIGH	Policy unverified across hazard catalogue
HZ-M01	Joint pinch / crushing in RUN	e	MED	Speed limits not bound to body regions
HZ-M04	Arm sweep strike on torso / head	e	MED	Speed reduction trigger not implemented
HZ-P02	Transient hand impact > 280 N	e	MED	PFL thresholds not declared
HZ-P03	Pinch finger between link and surface	e	MED	Cover gap not verified per ISO 13854

Top 10 gaps to close (of 28 total)

GAP-ID	Standard / clause	Description	Sev
GAP-15066-01	ISO/TS 15066 §5	PFL measurement campaign not performed	VHIGH
GAP-15066-02	ISO/TS 15066 A.2	Body-region thresholds not bound to joint torque caps	VHIGH
GAP-10218-02	ISO 10218-1 §5.5.3	E-Stop architecture undocumented	VHIGH
GAP-13849-02	ISO 13849-1 §7	MTTFD/DC/CCF figures absent	VHIGH
GAP-13849-03	ISO 13849-2	Validation plan & test report missing	VHIGH
GAP-13849-01	ISO 13849-1 §6	Cat 3 redundancy not implemented	VHIGH
GAP-LP-01	(this case)	LeRobot policy edge-case verification missing	HIGH
GAP-13854-01	ISO 13854	Joint pinch-zone gaps not analysed	HIGH
GAP-13857-01	ISO 13857	Separation-distance / safeguarding plan missing	HIGH
GAP-CYB-01	ISO 21434 (ref.)	Tele-op authentication scheme not documented	HIGH

Recommended next-revision actions

1	Run an ISO/TS 15066 PFL measurement campaign and publish body-region force/pressure data.
2	Specify and publish E-Stop architecture (Cat 1, dual-channel, link surveillance).
3	Compute and publish MTTFD/DC/CCF for each safety-rated channel (Damiao + Robstride).
4	Issue ISO 13849-2 validation plan and test report covering 31 safety functions.
5	Implement and document the mode FSM in firmware with mandatory E-Stop interlock at boot.
6	Verify cover gaps at all joint axes per ISO 13854.
7	Stand up an ISO 9001-aligned QMS subset (design review, NCR/CAR).
8	Verify LeRobot policy library against the 33-hazard catalogue; publish a coverage matrix.
9	Specify camera privacy posture (default-off, opt-in, recording LED).
10	Replace barrel-jack PSU connector with keyed XT60 / Anderson SB50.

Conclusion. Provisional acceptance for OE-1 (lab / desk supervised, fixtured) only. Closing the top 10 gaps unlocks OE-1 unconditionally and OE-2 conditionally. Deployment to OE-3/OE-4 requires the additional regulatory engagement documented in the safety case workbook tab 12.

Companion artefacts: [reBot_DevArm_Safety_Case.xlsx](#) · [reBot_DevArm_DFMEA.xlsx](#) · [reBot_DevArm_Requirements.xlsx](#) · [reBot_DevArm_Architecture.docx](#) · [reBot_DevArm_TestPlan.xlsx](#).